Alternative-protein spending in India and Pakistan

Drivers, conditional scenarios and the GFI comparison

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- India and Pakistan: what spending scenarios can establish
- Methods appendix: India–Pakistan spending scenarios
- Why these scenarios sit below GFI India
- Demand drivers, obstacles and proposed tests
- Does older sector optimism explain the GFI difference?
- Measurement contract and proposed harmonization
- Separate forecast-elicitation supplement
- Limitations and AI-use statements

India and Pakistan: what spending scenarios can establish

Review draft, 27 October 2026. AI-assisted. Human fact-checking and joint-report acceptance are still pending.

The question and the evidence

We wanted to know how spending on alternative-protein foods might grow in India and Pakistan to 2035, and what would make those paths more likely. No national record of repeat purchases exists for meat, dairy or egg alternatives in either country. Food accounts describe supply, surveys describe stated intentions, and business directories describe activity. We therefore built conditional spending scenarios and named the observations that would test them. Spending is the outcome. It does not measure adoption, protein intake or how much animal food is replaced.

Drivers and obstacles come first

Demand depends on a product reaching a buyer, fitting a meal and a budget, and being bought again. Eight factors organise the evidence. Each can push demand up or hold it back, and the balance differs by product, income group and place. The driver tables and evidence ledger keep India and Pakistan evidence separate.[1]

ID and factor	Pushes demand up	Holds demand back	What to watch
D01 Income	Rising incomes widen what households can buy.	Extra income may go to meat, dairy and other conventional foods.	Paid purchases by income group
D02 Affordability	A lower price per serving makes repeat purchase possible.	40.4% of Indians and 60.3% of Pakistanis cannot afford a healthy diet.	Retail price per serving and unsubsidised repeat purchase
D03 Dietary practice	Products that work in familiar dishes earn repeat use.	Taste, cooking method and doubts about processed food vary by group.	Repeated use in named meals
D04 Social legitimacy	Familiarity and trusted information reduce hesitation.	Favourable survey answers often do not become paid purchases.	First purchase and repurchase, reported separately
D05 Certification	A clear halal and food-safety route lowers buyer and seller risk.	General halal bodies do not settle a specific fermented product's status.	Product-specific assurance and authorisation decisions
D06 Public policy	India's BioE3 policy names smart proteins and R&D as aims.	Announcements may never reach funding, approval or shop shelves.	Dated records of implementation and uptake
D07 Capacity and access	Reliable production and distribution improve cost and availability.	Job adverts and directory entries do not prove capacity or a skills gap.	In-stock availability, unit cost and verified bottlenecks

ID and factor	Pushes demand up	Holds demand back	What to watch
D08 AI	Validated technical gains can cut cost or improve products.	Gains may stay in the lab or be matched by conventional producers.	AI versus non-AI cost and performance, then retail prices

These are eight organising factors. They are not eight measured effects. The Indian consumer survey behind D03 and D04 sampled 1,024 online respondents, skewed towards urban, affluent and educated people, and asked about hypothetical purchases. GFI India's talent study behind D07 drew on 20 industry and 19 academic respondents. Policy announcements under D06 are separate from implementation.[2][3][4]

What we did

We started from GFI India and Deloitte's estimate of India's 2022 market: INR 351 crore across plant-based meat, dairy and eggs. The source does not state its price basis. We treated it as 2022 retail spending, adjusted for inflation and converted it at the 2025 exchange rate. That gives about $45.3 million.[5]

We then grew India's spending per person with real income, urbanisation and an extra annual growth term. The extra growth term ranges from -2% (a recent US sales decline) to 17.8% (growth in Indian alternative-protein job adverts). For Pakistan, which has no national sales figure, we transferred India's path using relative income, urban share, animal-protein supply and industry activity. We drew four uncertain parameters 10,000 times. The plain-language walkthrough and equations show each step.[6]

Results and what moves them

Values are million US dollars at constant 2025 prices. Lower, middle and upper are the 10th, 50th and 90th percentiles of the 10,000 runs. They describe our assumptions, not calibrated odds of future sales.

Country	Year	Lower	Middle	Upper
India	2026	69	93	125
India	2030	102	188	334
India	2035	166	448	1,142
Pakistan	2026	1.5	3.6	6.6
Pakistan	2030	2.3	6.0	13.7
Pakistan	2035	3.6	11.8	36.1

Two sensitivities show how much the assumptions matter. Income growth and job-advert growth may partly measure the same thing. Removing that overlap cuts the extra growth ceiling from 17.8% to 6.0%. India's middle estimate then falls from $188 million to $120 million in 2030, and from $448 million to $215 million in 2035. For Pakistan, dropping the urban-share and animal-supply adjustments lowers the 2035 middle estimate from $11.8 million to $8.3 million. Pakistan's figures rest on a transfer from India that no Pakistani sales data has checked.[7]

Why the scenarios sit below GFI India

GFI's scenarios assume far faster growth than our model allows, and our US-based assumptions explain little of the difference. Our India 2030 middle estimate is $187.8 million. Raising the US-derived growth floor from -2% to 0% lifts it only to $202.2 million. With every parameter at its upper bound, the model reaches $462.0 million. GFI's lowest scenario converts to about $964.4 million.[8]

The difference comes from how each estimate is built. GFI sizes the market bottom-up by socio-economic group. It assumes plant-based products reach taste and price parity with conventional foods, helped by local supply chains, R&D and talent. It also expects whole categories to take off: plant-based eggs grow from 1 to 527 crore in the low case, and dairy from 250 to 4,827 crore. Our model scales one national figure by population, income and urbanisation, plus a capped extra growth term. It has no price-parity term and no term for new categories. To reach GFI's low case, that extra growth would need to be 29% to 36% a year. Our upper additional-growth assumption is 17.8%.

GFI domestic scenario	Reported 2030 value, INR crore	Multiple of reported 2022 total	Extra annual growth our equation would require
Low	7,475	21.30x	29.2-35.7%
Medium	12,779	36.41x	38.1-45.1%
High	18,963	54.03x	45.1-52.4%

The required-growth range varies income elasticity across our existing bounds. The reviewer's four questions have these answers:

- **Are GFI's figures more optimistic because they are older?** Possibly, but we cannot confirm it. GFI used a 2022 baseline. Dated 2021 and 2022 sources show both upbeat sector messaging and slowing US sales, so age alone does not explain the assumptions.
- **Does GFI include factors we do not?** Yes. GFI's scenarios rest on price parity, new-category entry and supply-chain localisation. Our model has no separate terms for them, so we cannot say how much each adds.
- **Is projecting from the US wrong?** The US is a minor factor here. Our starting figure is Indian. US data set only the growth floor and the income elasticity ceiling. Moving the floor changes India's 2030 estimate by 7.7%. Fixing elasticity at 1 lowers it to $156.0 million.
- **What else could matter?** GFI does not state its price basis, so its growth rates cannot be labelled nominal or real. Category scope also differs slightly: fermentation-derived meat is 4.3% of GFI's low case.

The overlap sensitivity points the other way and widens the gap. Neither forecast is validated by the comparison. The disagreement does identify the assumption to test first: whether Indian buyers will reach price and taste parity fast enough to sustain growth above 30% a year. Proposals R01 and R02 below measure that directly. The GFI comparison gives the full category and factor tables.[9]

US benchmark and the separate elicitation

The 2035 middle scenarios equal about $0.28 per person in India and $0.040 in Pakistan. That is roughly 1.22% and 0.17% of the fixed US 2025 plant-based retail benchmark per person. These are per-person comparisons with a broader US product basket. They are not shares of the US market or comparisons with a US 2035 forecast.[10][11]

A separate panel of 60 AI-generated forecasts gives a second view. Its regional headline excludes fermentation, and its spread describes differences between runs, not sales uncertainty. GFI's figures fed both our model and the panel's evidence packets, so agreement between them is not independent confirmation. A supplementary US run is reported separately and does not stand in for the original US panel.[12]

What follows from the evidence

This chapter shows which assumptions produce the country estimates and which observations would challenge them. Each recommendation below answers one or more drivers from the table above. Each is a test with a named measure. The recommendation register lists the proposed actor, comparison and outcome for each.[13]

- **R01 Retail price, stock and purchase audit (D01, D02, D07).** Retailers and a research lead record prices per serving, stock and actual purchases for matched products in both countries. This tests the price-parity assumption behind GFI's faster growth.
- **R02 Meal-fit and repeat-purchase study (D02, D03, D04).** A local study team tests one product against a familiar dish. It tracks paid trial, repeat purchase after discounts end, and any reduction in animal foods.
- **R03 Fermentation product dossier (D05, D06).** A product owner, with halal and food-safety advisers, documents ingredients and process. The output is a product-specific halal and approval decision before any sale.
- **R04 and R05 Technical trials (D07).** In India, an employer names one production bottleneck and tests a fix against current performance. In Pakistan, a first check confirms that facilities, staff and inputs can support an ingredient trial.
- **R06 AI versus non-AI comparison (D07, D08).** A technical team compares cost and output with and without AI tools. A second step checks whether any saving reaches shop prices.[14]
- **R07 Policy implementation register (D05, D06).** A dated register tracks each policy from announcement to funding, approval and market effect. Missing steps stay recorded as missing.

Methods appendix: India–Pakistan spending scenarios

AI-assisted review draft, 27 October 2026. Human factual/code verification pending. Proposed joint-report methods contribution; definitions are proposed harmonization, not a recorded team agreement.

Read this first: the calculation in plain language

We ask how domestic spending on alternative-protein foods might develop in India and Pakistan in 2026, 2030 and 2035. We do not measure national adoption or predict how much conventional meat will be replaced. The following steps explain the retained model; the exact equations below are unchanged.

1. **Start with India.** GFI India/Deloitte estimates a 2022 market of INR 351 crore for plant-based meat, dairy and eggs. We assume a 2022 price basis and retail valuation, then convert it to 2025-price dollars: about $45.3 million. These interpretations are assumptions because the source does not specify them.
2. **Carry spending per person forward.** Income growth and urbanisation raise or lower the path. An additional annual growth term represents other changes imperfectly; its bounds come from a US sales decline and Indian vacancy growth. These are chosen proxies, not relationships fitted to purchases.
3. **Construct Pakistan's conditional path.** There is no equivalent verified national sales anchor in the retained evidence. We transfer India's per-person path using relative income, urban share, animal-protein supply and industry-activity proxies, then multiply by Pakistan's population. This adds uncertainty and does not establish observed Pakistani sales.
4. **Repeat under different assumptions.** The model draws four parameters 10,000 times. Each draw is reused across countries and years. The lower, middle and upper values are the 10th, 50th and 90th percentiles of those calculations; they are not calibrated probabilities of future sales.
5. **Change assumptions deliberately.** Retained sensitivities test overlapping income/industry growth and Pakistan multipliers. The new GFI companion also changes the US-derived growth floor and elasticity assumptions separately and calculates the extra growth required to reach GFI's targets. It leaves the historical reference results unchanged.
6. **Interpret the limits.** GFI's domestic scenarios, our assumption percentiles and repeated AI responses are different objects. Shared GFI inputs prevent treating their agreement as independent confirmation. The main empirical need is comparable sales and repeat-purchase evidence, especially for Pakistan.

For the numerical comparison, see Why these scenarios sit below GFI India. Drivers and proposed tests explain the mechanisms that the aggregate equation cannot identify separately.

1. Question, estimand and evidence classes

The calculation explores domestic alternative-protein spending in India and Pakistan in 2026, 2030 and 2035. Its intended scope is plant-based replacement foods and fermentation-derived human foods, excluding cultivated meat. The 2022 Indian anchor contains plant-based meat, dairy and eggs; the source's

forward domestic categories include fermentation-derived meat. The model carries one aggregate forward and does not separately estimate entry of every proposed product category. Its interpretation must retain that limitation.

The proposed valuation is retail spending in market-exchange-rate US dollars at constant 2025 prices. **Retail valuation and the Indian source's nominal-2022 price basis are assumptions.** They are not explicit properties established by the original GFI/Deloitte summary. US coverage is broader and includes prepared meals. These differences prevent a fully matched cross-country market comparison.

Observed macro quantities, published external market estimates, analyst scenarios, model-elicited projections and missing evidence are recorded as different classes. The proxy scenarios estimate neither adoption probabilities nor physical protein volumes. FAOSTAT supply is an input proxy, not household consumption or alternative-protein demand. No model output is converted into an observed baseline.

The model was not fitted to a time series of alternative-protein purchases, and its future outcomes are unresolved. Calibration, predictive accuracy and probability coverage have not been demonstrated. Country-specific inference is limited to India and Pakistan.

2. Source selection and numerical inputs

The retained model reads **106 input rows** in `data/altprotein_proxy_inputs.csv`. Each row has an input ID, country/year, quantity, unit, source ID, tier, raw locator, capture reference/hash where available, and notes. The source collection is purposive; it was not a preregistered systematic review. Existing acquisition logs record unsuccessful and incomplete retrieval separately. Source dates and collection dates must not be conflated.

Input family	Source/vintage and retained use	Main limitation
Indian market anchor and external scenarios	GFI India/Deloitte, 2022-context analysis, final-page domestic table: 351 crore in 2022; 2030 low/medium/high 7,475/12,779/18,963 crore	Published model estimates; price/valuation basis unstated; 2030 scenarios are not observations or probability bounds.
US sales benchmark	GFI/SPINS 2025 plant-based retail sales, $7.9B; reported annual dollar change −2%	Broad US categories; nominal annual sales change is repurposed as a real non-income drift bound by assumption.
Population, GDP growth, inflation	IMF WEO October 2026: LP, NGDP_RPCH, PCPIPCH	Mixed historical estimates/outturns and projections; published horizon ends in 2031.
Urban share	WDI SP.URB.TOTL.IN.ZS, 2015–2025	National urban definitions differ; future trend is extrapolated.
Exchange rate and income levels	WDI PA.NUS.FCRF; NY.GDP.PCAP.CN/CD	Market exchange rates, not PPP; observation years differ in the US–India elasticity construction.
Animal-protein supply	FAOSTAT 2023 protein supply, seven animal-food groups	Supply balance, not individual intake, willingness or replacement-product demand.
Industry activity	AltProtein.Jobs directory described as 2024 and annual vacancies 2023–2025	Job-board coverage and country assignment are imperfect; directory records may overlap; not employment or skill shortages.

The source ledger and source-specific rights should accompany extracted facts. Reproduction from the numeric input CSV does not require redistributing source workbooks, full PDFs, cached web pages or private team documents.

3. Exact implemented transformations

Notation: country `c` is India `I` or Pakistan `P`; year is `t`; `P_c(t)` denotes population **in millions**, `U_c(t)` urban share **in percent**, `I_c(t)` real GDP per person relative to 2022, `s_c(t)` annual dollars per person, and `M_c(t)` total spending **in million dollars**. Percentage units cancel in urban-share ratios.

3.1 Price conversion and anchors

Let `F = (1 + 5.4/100)(1 + 4.6/100)(1 + 2.0/100)`, using Indian consumer-price inflation in 2023–2025. Let `X = 87.158448` INR per USD, the retained rounded WDI 2025 period-average exchange rate.

```
Indian anchor A = 351 × 10 × F / X = 45.286640 million USD at 2025 prices
s_I(2022) = A / P_I(2022) = 0.031771 USD per person
s_US(2025) = 7.9 × 1,000 / P_US(2025) = 23.106847 USD per person
```

The factor 10 converts crore rupees to million rupees. The US 2025 dollars already use the selected price year. The Indian anchor is **a 2022 observation-year estimate converted to 2025 prices**, not a 2025 market observation.

3.2 Population and real-income paths

The retained IMF population series is used through 2031. Beyond it:

```
P_c(t) = P_c(2031) × [P_c(2031)/P_c(2030)]^(t−2031), for t > 2031
I_c(2022) = 1
I_c(t) = product over y=2023…t of [(1 + g_c(min(y,2031))/100)
                                    / (P_c(y)/P_c(y−1))]
```

Here `g` is real GDP growth. Holding the 2031 GDP and population growth rates through 2035 is an analyst extrapolation, not an additional IMF forecast. Retained 2035 income indices are 2.167826 for India and 1.254629 for Pakistan.

For each country, ordinary least squares on the eleven annual urban-share observations from 2015 through 2025 supplies a slope `b_c`. Future urban share is `U_c(2025) + b_c(t−2025)`. The function uses observed values through 2025. No estimated uncertainty or cross-country harmonization is added. Retained 2035 shares are 38.831347% and 43.289498%.

3.3 Parameter bounds

Four parameters are independently sampled from uniform distributions once per draw. The **same sampled parameters are reused across years and countries** within that draw; country/year results are therefore related, not independent samples.

Parameter	Distribution/bounds	Construction and interpretation
Income elasticity `e`	Uniform(1, 1.810389…)	Lower bound assumes proportional income scaling. Upper bound attributes the entire US–India per-person spending gap to the market-dollar income gap. It is not an estimated causal elasticity.
Additional annual drift `d`	Uniform(−0.02, 0.17832570…)	Lower bound is US 2025 nominal retail dollar decline; upper bound is `(529/381)^(1/2)−1`, Indian vacancy CAGR for 2023–2025. Mapping these into future spending is an assumption.
Animal-supply weight `w`	Uniform(0,1)	Applies none to all of Pakistan's animal-supply ratio multiplicatively.
Pakistan industry factor `k`	Uniform(0.102303…,0.667385…)	Bounds are Pakistan/India directory-row density and vacancy density, both divided by 2025 population.

The full-precision values are computed by the retained code from inputs, not from the rounded bounds above. Directory counts are 119 for India and 2 for Pakistan. 2025 vacancies are 529 and 58. Neither defines a complete national industry census.

For the upper elasticity bound, India's 2022 GDP per person in local currency is inflated by `F` and divided by `X`. US 2025 GDP per person is already in market dollars at 2025 prices. Then:

```
e_max = log[s_US(2025)/s_I(2022)]
        / log[GDPpc_US(2025)/(GDPpc_I(2022, INR) × F/X)]
```

The retained spending and income ratios are 727.301293 and 38.081810. They have a common price basis but different observation years. A same-price-basis conversion does not make them a matched contemporaneous comparison. Urbanisation, tastes, availability, category coverage and measurement also contribute to the observed gap; attributing it entirely to income is only a bounding assumption.

3.4 India and Pakistan equations

```
s_I(t) = s_I(2022) × I_I(t)^e × [U_I(t)/U_I(2022)] × (1+d)^(t−2022)

R_income(t) = [GDPpc_P(2025)/GDPpc_I(2025)]
              × [I_P(t)/I_P(2025)] / [I_I(t)/I_I(2025)]

s_P(t) = s_I(t) × R_income(t)^e × [U_P(t)/U_I(t)] × R_animal^w × k

M_c(t) = s_c(t) × P_c(t)
US comparison percentage_c(t) = 100 × s_c(t)/s_US(2025)
```

The 2025 Pakistan/India income ratio is 0.590535. `R_animal = 31.52/19.67 = 1.602440…`, from the sum of bovine meat, eggs, fish/seafood, milk excluding butter, mutton/goat, pigmeat and poultry protein supply. The industry and dietary multipliers are not independently identified effects. Their multiplication may double-count correlated conditions.

No Pakistan retail observation is used to fit or validate the transfer equation. Missing baseline evidence is never assigned a zero. No separate AI or policy multiplier is applied. No cultivated-meat entry term is present.

4. Draws, aggregation and output

The implementation uses Python's standard-library pseudorandom generator, `random.Random(20261015)`, and 10,000 draws. Every iteration draws `e`, `d`, `w`, `k` in that order. Results for each country/year are sorted. For quantile fraction `q`, compute `h=(n−1)q`, `i=floor(h)`, `j=ceil(h)`, and return `x[i]+(x[j]−x[i])(h−i)` with zero-based indexing. Output CSV values are formatted to six decimal places.

The main table reports q=.1, .5 and .9. They describe variation under chosen parameter distributions. They are **not** confidence bounds, a calibrated 80% predictive interval, or empirical household uncertainty. Display rounding must not imply greater accuracy. The chapter uses $448m and $11.8m for the retained 2035 medians 447.504341 and 11.759263.

5. Retained sensitivities and external benchmark

Sensitivities reset the same seed, retaining paired draws. Even a fixed or unused parameter is drawn before being replaced; this preserves pairing.

1. **Income/drift overlap:** annual real income growth from 2023 to 2025 is 6.548202%. The alternative upper drift is `d_max − income_growth × e_max = 5.977774%`; the lower bound stays −2%.

India's 2035 p10/p50/p90 become 134.658168 / 215.060659 / 341.253412 million; Pakistan's become 2.382182 / 5.964900 / 11.690860 million. This is a sensitivity, not a fitted correction.

2. **Pakistan overlap:** set its relative urban share and animal-supply multipliers to 1 while retaining India's path, income ratio and industry factor. Pakistan's 2035 p10/p50/p90 become 2.575709 / 8.318219 / 25.341349 million.
3. **One parameter fixed:** hold each parameter at its lower and upper bound while drawing the others. India's 2035 median ranges are 332.438965–622.332639 million for elasticity and 127.357046–1398.084812 for drift. Pakistan's corresponding pairs are 12.293369 / 11.255477 for elasticity, 3.652403 / 40.094912 for drift, 9.273218 / 14.859778 for animal weight, and 3.468582 / 22.627744 for industry factor. These are pairs of conditional medians, not uncertainty intervals or variance decompositions. Higher elasticity need not increase Pakistan's transfer estimate because its relative income factor is below one.

The existing comparison converts GFI India's 2030 external scenarios using the **same 2022-to-2025 inflation and exchange-rate transformation** as the anchor, yielding 964.437694 / 1648.769136 / 2446.639731 million. This assumes the external scenarios share that source price base; the summary does not establish it. The fact that all retained proxy draws fall below GFI's low case is a conditional disagreement, not a validation test or proof of which forecast is correct. Preserve source rupee values alongside any converted comparison until the price basis is resolved.

6. Behaviour, policy and talent evidence

Behavioural evidence is read for the actual question and sampled population. Stated willingness, paid first purchase, unsubsidized repeat purchase and animal-food displacement are separate outcomes. The selective nine-study extraction package has no India finding and is not a systematic review of national adoption; the bilateral discussion draws on additional studies. Do not turn "not measured in these sources" into "does not occur."

Policy sources establish the dated instrument or announcement and its stated jurisdiction. Implementation, disbursement, product authorization and uptake need different evidence. Halal status and food authorization are distinct questions. This appendix does not provide a legal determination or assume a product's certification from general institutional powers.

Talent evidence separates job-board advertisements, directory entries, sampled employer testimony and national workforce measures. GFI India's study's 20 industry responses and 19 academic participants are study cohorts. Its findings may motivate a pilot; they do not identify a nationally optimal training portfolio. Recommendations in the chapter are author proposals that require baseline and outcome checks.

7. Validation, reproducibility and remaining holds

Run `python scripts/reproduce.py` from the package root. The packaged engine `scripts/build_altprotein_proxy.py` reads the 106 aggregate input rows and writes `build/proxy_scenarios.csv`. Its private acquisition/transcription mode has been removed; the mathematical engine, random draw order and output format are preserved. No source acquisition or model call is required.

The 22 October package verification regenerated all 127 rows byte for byte against `data/reference_proxy_scenarios.csv`. Independent denominator and sensitivity checks are included in `tests/test_reproduction.py`. Clean-folder verification is recorded in the release receipt. Reproduction establishes consistency with the specified assumptions, not empirical validity.

Human reviewers must resolve the Indian price/retail basis, judge the crosswalk and proxy assumptions, verify included sources/data/code, and sign off the final claims. A new national sales series should be documented as new evidence and evaluated explicitly rather than silently replacing the original anchor. The separate supply benchmark's out-of-sample performance does not validate alternative-protein spending. Full source/capture pointers and rights limits are in claim map and source ledger.

Why these scenarios sit below GFI India

Local review candidate, 27 October 2026. Numerical diagnostics; human factual/code verification pending.

The retained India 2030 median is $187.8 million, compared with $964.4 million for GFI's low domestic scenario under the existing conditional price conversion. This appendix asks which parts of that disagreement are demonstrated by the equations and which remain hypotheses. No historical headline is replaced. The companion `scripts/gfi_gap.py` writes its results separately in `build/gfi-gap/`.

Compare the same question first

The comparison is India, domestic spending, 2030. Exports, economic impact and employment are excluded. GFI's source reports rupee amounts for plant-based meat, dairy, eggs and fermentation-derived meat. Its price and retail-valuation bases are unstated. Our retained conversion assumes a common 2022 price basis, inflates to 2025 and converts at the 2025 exchange rate. Dollar comparisons below depend on that assumption. The original rupee amounts remain visible so that an unresolved definition is not hidden by conversion.[15]

GFI's low/medium/high cases are scenarios; our P10/P50/P90 values are percentiles under chosen parameter distributions. They are not equivalent statistics. Our aggregate intends to include fermentation, but has no separately estimated category-entry amount. The regional AI elicitation excludes fermentation. These distinctions limit both category attribution and comparisons between result families.

Category decomposition

The table transcribes physical PDF page 4. Each 2030 cell gives the reported value in INR crore and the annual growth implied by its 2022 base over eight years. The price basis is unresolved, so these growth rates are not labelled nominal or real. The source's fermentation dash stays undefined; neither its CAGR nor absolute increase can be calculated from a known base.

Category	2022, INR crore	Low 2030: value (growth/year)	Medium 2030: value (growth/year)	High 2030: value (growth/year)
Plant-based meat	100	1,803 (43.55%)	3,525 (56.10%)	5,884 (66.42%)
Plant-based dairy	250	4,827 (44.78%)	7,643 (53.34%)	10,625 (59.79%)
Plant-based eggs	1	527 (118.89%)	989 (136.81%)	1,416 (147.68%)
Fermentation-derived meat	Source dash; undefined	318 (N/A)	622 (N/A)	1,038 (N/A)
Overall domestic market	351	7,475 (46.57%)	12,779 (56.73%)	18,963 (64.66%)

All three 2030 category totals reconcile. The known 2022 entries sum to the reported 351 crore, but this does not turn the dash into an observed zero. Eggs' large CAGR reflects its one-crore base: the low-case absolute increase is 526 crore, compared with 4,577 crore for dairy. Absolute changes, full precision and source locators are retained in the category-growth table and source extraction.

Fermentation is **4.254% of GFI's low-case total**. Subtracting its 318 crore from that target reduces the difference between the conditionally converted GFI low case and our median by **5.283%**. The denominators differ. This arithmetic exercise cannot isolate how much fermentation explains the forecasts' disagreement: our own aggregate already intends to include it, without a category-specific estimate. The target comparison reports both denominators for all three scenarios.

Which model factors produce the Indian path?

For any one draw, the spending multiple is exactly:

```
M_India(2030) / A = population_ratio × income_index^elasticity
                  × urban_share_ratio × (1 + extra_growth)^8
```

Here A is the converted 2022 anchor, not a 2025 observation. The following illustration fixes elasticity and extra growth at their parameter midpoints. It is not the median of the simulated totals.

Factor, 2022 to 2030	Multiplier	Interpretation
Population	1.069955	More people at a given spending per person
Real income raised to elasticity	1.968639	Chosen spending response to the real-income path
Urban share	1.072066	Proportional urban-share scaling
Additional growth	1.839483	Eight years of the midpoint extra-growth assumption
Product	4.153832	$45.286640m anchor becomes $188.113110m
GFI reported low/medium/high multiples	21.296 / 36.407 / 54.026	Reported 2030 amounts divided by 351 crore

The retained simulated median is $187.811844 million. A product of marginal factor medians need not recover it. The companion preserves each draw's factors and exact product; additive log factors describe the same arithmetic and do not identify causal contributions. The complete draw file is available in the offline reproduction package, not loaded into the website's interactive data payload.

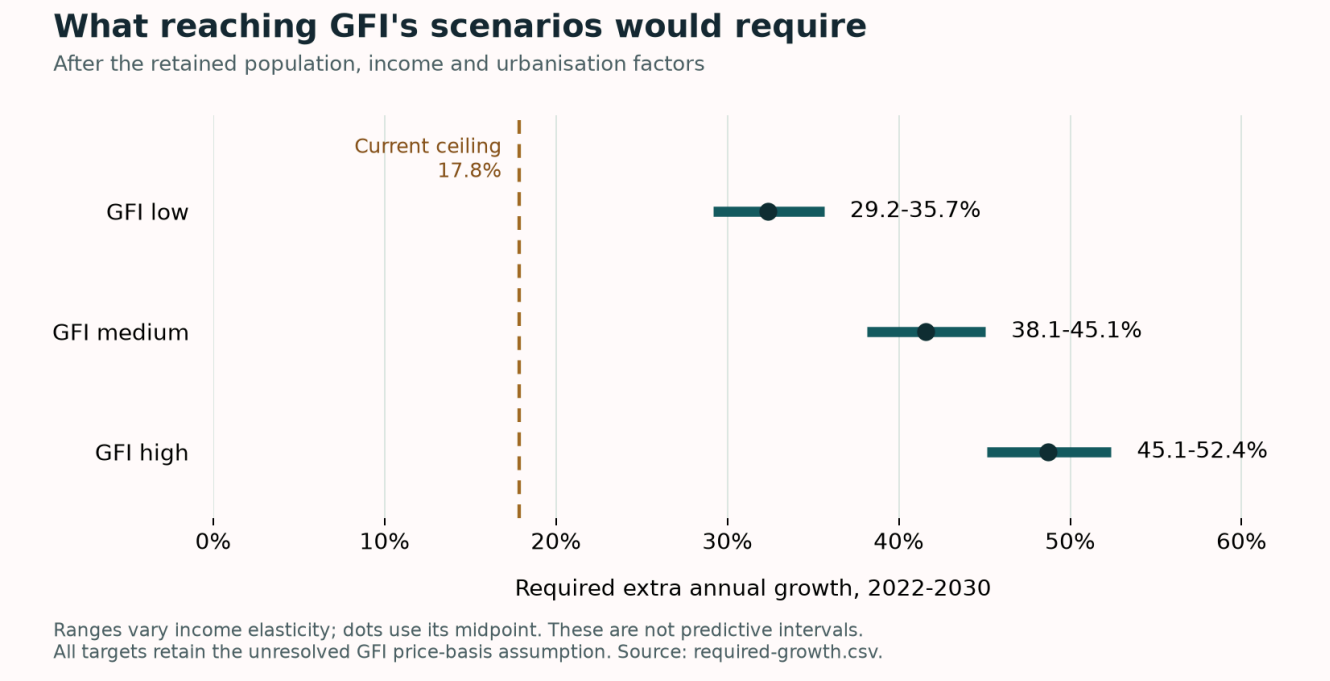
The fastest sampled draw grows by 33.556509% annually. Evaluating the parameter upper bounds gives an analytical maximum of 33.686486%, or $462.029988 million in 2030. Even that boundary lies below the conditionally converted GFI low case. This is a property of the selected model bounds. It does not establish that GFI is wrong or that our model is calibrated.

What growth would reaching GFI require?

We solve the same equation backwards, holding each target and the other factors fixed:

```
d_required(e, T) = [T / (A × population_ratio × income_index^e × urban_ratio)]^(1/8) − 1
```

The target T and anchor A must share the stated price basis. The figure varies income elasticity across the retained bounds. The dot uses midpoint elasticity; the range is assumption sensitivity, not a predictive interval.



Extra annual growth required to reach GFI: low 29.2-35.7%, medium 38.1-45.1%, high 45.1-52.4%, compared with the retained 17.8% ceiling.

Target	Required extra growth at midpoint elasticity	Range over elasticity assumptions
Low	32.38% per year	29.19-35.65%
Medium	41.56% per year	38.14-45.05%
High	48.72% per year	45.13-52.39%

The current upper extra-growth assumption, 17.832570%, comes from Indian job-advertisement growth. It is not an estimated ceiling on purchases. GFI matching would require changing the assumptions; doing so solely to hit the target would add no independent evidence. The required-growth grid, figure data and SVG figure accompany the calculation.

Testing the US-derived assumptions separately

US evidence sets an upper elasticity bound and a lower extra-growth bound, as well as providing a descriptive 2025 spending benchmark. A change to the growth floor alone therefore does not remove US influence. The companion consumes the same four uniform draws in the same order for every case; it rescales the growth draw into a new interval rather than clipping negative values to zero.

Case	India 2030 median, $m	Pakistan 2030 median, $m
Retained baseline	187.812	5.977
Extra-growth floor 0% instead of -2%	202.214	6.504
Elasticity fixed at 1	156.034	6.566
Both changes	167.830	7.143

The zero-floor change raises India's median by 7.668%; it does not approach GFI's low case. Fixing elasticity at 1 lowers India while raising Pakistan, because Pakistan's relative-income multiplier is below one. These are diagnostic alternatives, not empirically estimated South Asian corrections. The full grid, all years, both countries and distribution summaries are in sensitivity-summary.csv.

The retained overlap sensitivity lowers India's 2030 median from $187.8 million to $119.6 million, widening the same-year GFI discrepancy. Its 2035 median also falls, from $448 million to $215 million. Pakistan's existing urban/animal omission and industry-factor sensitivities are reproduced. Its industry bounds use incomplete directory and vacancy measures; the lower directory bound includes only two Pakistan records. No national Pakistan sales series fits or validates the transfer. Explaining India's disagreement with GFI cannot resolve that separate evidential gap.

Four questions raised by the comparison

Are older data responsible? The GFI summary uses a 2022 baseline/context, but its exact publication and model-development dates are unresolved. Dated 2021-22 US releases contain both upbeat sector messaging and evidence of slowing growth. They establish context, not why Deloitte selected particular assumptions. The bounded dated-source review therefore leaves the causal explanation untested.

Does GFI include factors our model lacks? Its methodology explicitly discusses socioeconomic-group penetration, adoption, replacement, relative conventional/plant-based prices, taste and price parity, supply-chain localization, R&D and talent investment. Page 2 attributes scenario construction to GFI's target-market research and Deloitte's proprietary market-sizing, with expert interviews informing taste/price parity. Our aggregate model has no separately identified price-parity or category-entry term. Residual growth may implicitly reflect some of these mechanisms; their individual contributions cannot be recovered from the equation. This is a documented difference in representation, not an additive attribution of the numerical gap.[16]

Is projection from the US inappropriate? Our starting market estimate is Indian, while two parameter bounds use US evidence. The paired tests above show their numerical influence separately. The original US panel's politics/health/R&D themes remain reviewer-supplied prompts for the country-specific

driver comparison, pending verified provenance of that study. A US trend cannot establish the same mechanism or effect size in India or Pakistan.

What else could matter? Source price/retail bases, observation years, product/channel scope, scenario definitions, chosen parameter distributions and proxy validity remain material. GFI supplies our baseline and appeared with forward estimates in the regional elicitation packets. Neither matching GFI nor agreement with those responses is independent validation. GFI's public summary does not disclose sufficient detailed proprietary parameters to calculate a definitive causal decomposition.

Reproduction and review boundaries

From the offline package, run `python scripts/reproduce.py` and `python scripts/gfi_gap.py`, followed by the unit suite. The numerical companion reproduces 55 overlapping values at the original six-decimal precision; the retained engine still reproduces all 127 reference rows byte for byte. Its new output files are separate. Tests cover source missingness, category reconciliation, hand-calculable growth examples, target inversion, exact draw factors, paired random-number ordering, monotonic floor changes, bounds and deterministic repeatability.

These checks establish implementation consistency. Source-specific human factual/code review, scientific acceptance and rights decisions remain pending. Results may be inspected locally without implying that a release or joint-report inclusion has been approved.

Demand drivers, obstacles and proposed tests

Local review candidate, 27 October 2026. Human factual, code and editorial review pending.

These eight factors organize the argument. They are not an exhaustive census of demand drivers or eight estimated causal effects. A possible mechanism, an observed association, a policy announcement and a model assumption support different claims. India and Pakistan evidence remains separate, and missing Pakistan evidence is not filled with an Indian or US estimate.

The detailed evidence ledger records source dates and locators, geography, population/product scope, evidence class, model treatment, limitations and review status. The recommendation register specifies proposed actors, comparators, outcomes and follow-up periods. No partner, budget or programme is represented as committed.

Mechanisms and obstacles

ID / factor	What could increase demand	What could constrain or reverse it
D01 Income	More purchasing power can expand feasible purchases.	Additional income may go to conventional foods; income distribution and actual purchase responses are unmeasured.
D02 Affordability	Lower real price per comparable usable serving can improve access.	Parity with meat does not necessarily imply affordability against other foods or household budgets.
D03 Dietary practice	Products that perform well in familiar meals may earn repeat use.	Taste, preparation, cuisine and health/processing perceptions vary by product and group. Perceptions are not clinical evidence.
D04 Social legitimacy	Familiarity and trusted information may reduce uncertainty.	Favourable hypothetical answers can coexist with distrust, stigma or low paid repeat purchase.
D05 Certification	A documented product-specific assurance route can reduce uncertainty.	Halal assessment and food authorization are distinct; neither follows from a general institution alone.
D06 Public policy	Implemented support can improve capabilities or entry conditions.	Announcements may not become disbursement, usable infrastructure, authorized products or purchases.
D07 Industry capacity and access	Verified technical capabilities and dependable distribution can improve performance, cost and availability.	Advertisements and directory entries do not establish skill shortages, capacity, product quality or repeated retail access.
D08 AI	Validated technical gains may improve production performance or cost.	Scale-up, validation costs and retail pass-through can absorb gains; conventional producers may improve too.

The net direction of D03–D08 is conditional or mixed. No magnitude or country ranking is inferred from this table.

Country evidence and observations that would change the assessment

ID	India evidence	Pakistan evidence	Useful signpost
D01	WEO income path is a macro input; spending elasticity is assumed.[17]	Relative-income transfer has no fitted national alternative-protein sales anchor.	Actual purchases by defined income group, alongside macro revisions.
D02	SOFI estimates 40.4% unable to afford a healthy diet in 2024.[18]	The corresponding estimate is 60.3%; neither number measures an alternative-product premium.	Dated matched portions, retail prices, stock and paid repeat purchases.
D03	An online intention study and GFI's modelled category mix provide limited product context.[19][20]	Food-balance supply cannot establish preferences for chicken or milk replacements.[21]	Meal-specific choices, preparation success and repeat use by defined groups.
D04	Familiarity correlates with acceptance in Bryant et al.'s selected sample; this is not an intervention effect.[22]	This retained package provides no matched national paid/repeat-purchase measure.	Separate intention, first paid purchase, repeat purchase and displacement.
D05	This bounded source set establishes no product-specific determination for the proposed fermentation example.	The 2016 Act establishes institutional provisions, not a named product's status.[23]	Written ingredient/process-specific assurance and authorization decisions.
D06	BioE3 identifies smart proteins, R&D and Bio-AI/biomanufacturing infrastructure as policy aims.[24]	Halal institutional provisions do not establish equivalent smart-protein funding or uptake.	Separate records of announcement, implementation, funding, approval and market outcomes.
D07	Talent-study cohorts and vacancy records identify issues to investigate.[25][26]	Two directory rows and 58 recorded 2025 vacancies are coverage-limited proxies, not a census.	Verified staff/facilities, process performance, unit cost and outlet availability.
D08	BioE3 supplies a policy-intent reference, not an estimated AI spending effect.[27]	No national AI effect is established in this retained specification; that does not show applications are absent.	AI/non-AI technical performance and cost, then separately observed retail outcomes.

The Indian survey contains 1,024 online respondents and is skewed urban, affluent and educated. It has no Pakistan sample. Its hypothetical willingness must not be relabelled as national paid demand. SOFI measures economic access to a healthy diet, not protein deficiency or the affordability of a particular replacement product. FAOSTAT food balances measure supply, not individual intake or substitution.

What the current model represents

D01 enters explicitly through real income and a chosen elasticity. D03 enters Pakistan's transfer only through an animal-protein-supply multiplier, which does not separately represent taste or dietary choice. D07 supplies the vacancy-growth ceiling for additional growth and the relative industry factor for Pakistan.

Their conversion into spending is assumed, not fitted to purchases.

D02 and D04–D06 have no separately identifiable coefficients. They may be among the mechanisms implicit in additional growth, but the aggregate model cannot quantify their separate effects. D08 has no AI multiplier. Adding an AI or policy bonus would require a new justification and a check for overlap with existing growth assumptions. Cultivated meat is excluded. Fermentation is in the proxy's intended scope but has no separately estimated amount. These are properties of the retained specification, not evidence that the mechanisms do not matter.[28]

US transferability questions

The three themes below were supplied in reviewer feedback. The original US-panel passage and executed methods have not been established in this revision. They therefore remain **comparison prompts**, not newly verified original-panel findings. Umar's supplementary US arm is a separate exercise; its questionnaire provenance does not establish equivalence with another panel. No private panel quotation or questionnaire is reproduced here.

Reviewer-supplied theme	Relevant driver IDs	Country-specific question
Political environment	D04–D06	Which dated policies, assurance rules or social conditions apply to the named product, and which implementation stages are evidenced?
Health trends	D02–D04	Which buyers hold which beliefs about a specific product, and do those beliefs predict paid repeat purchase at its actual price?
R&D landscape	D07–D08	Which technical constraint affects product performance, cost or availability locally, and does resolving it improve a measured outcome?

The US sales series also has three mathematical roles in the retained analysis: an elasticity bound, an additional-growth lower bound and a historical per-person comparison. The separately reported sensitivities test those assumptions. A US sales change cannot, by itself, assign signs or effect sizes to the themes above.

Recommendations follow the evidence and its gaps

These are proposed learning activities. They are not a ranking of expected returns or an existing implementation programme. Every major obstacle has a disposition: measure it, run a bounded feasible test, seek qualified clarification, or monitor a specific stage.

Recommendation	Driver IDs	Proposed action and outcome
R01 Comparable retail audit	D01, D02, D07	Track matched prices, stock and actual purchases separately in each country; an unavailable sales series remains unavailable.
R02 Meal-fit and purchase study	D02, D03, D04	Test a defined product and meal; distinguish paid trial, unsubsidized repeat purchase and separately measured displacement.
R03 Fermentation product dossier	D05, D06	Specify ingredient, process, product and jurisdiction; obtain qualified assurance and food-authorization clarification separately before a pilot sale.
R04 India technical trial	D07	Confirm one employer-nominated constraint and compare performance and cost against a credible baseline.
R05 Pakistan feasibility assessment	D07	Verify facilities, staff, inputs and product need before deciding whether an ingredient-application trial is feasible.
R06 AI/non-AI comparison	D07, D08	Measure usable output, consistency and total unit cost against a credible non-AI process; assess retail pass-through separately.
R07 Policy stage register	D05, D06	Track documented implementation and product decisions; do not convert an announcement into a growth bonus.

The register's 8–12-week purchase follow-ups, 3–6-month technical pilots and quarterly policy reviews are proposed design horizons, not empirical findings or booked activities. Feasibility, sample, comparator, cost ceiling and decision threshold must be set before a study begins. Current evidence does not justify numeric success thresholds. Product-specific assurance questions must be resolved before any proposed sale, regardless of a nominal study timetable.

Does older sector optimism explain the GFI difference?

Bounded retained-source review, 27 October 2026. Local review candidate; human verification pending.

The retained sources support a limited conclusion: upbeat US sector messaging existed in 2021–22, alongside evidence of a slowdown. They do **not** identify the cause of GFI India/Deloitte's scenario assumptions. An older-information explanation remains a plausible, untested hypothesis, not the demonstrated reason for the numerical gap.

Evidence for and against the hypothesis

Source and observation period	Relevant finding	What it establishes
GFI/PBFA release, 6 April 2021; US sales in 2020	Reported 27% growth in plant-based retail dollar sales to $7 billion and presented strong expectations for continuing growth.[29]	A dated example of US sector optimism. The release also identified a pandemic retail boost; the observation is not an India growth rate.
GFI/PBFA release, 24 March 2022; US sales in 2021	Reported 6.2% aggregate dollar-sales growth to $7.4 billion and discussed a possible narrowing price gap.[30]	Continued positive messaging, but the same release reported plant-based meat repeating $1.4 billion in sales and described inflation and supply disruption. This is contrary evidence against a uniformly accelerating market.
GFI India/Deloitte economic summary; 2022 baseline and 2030 scenarios	Describes socioeconomic segmentation, adoption, replacement, prices, localization, R&D and talent assumptions.[31]	Documents intended model mechanisms. The public summary does not disclose enough numerical assumptions or dating to attribute its scenario growth to US optimism.

The two press releases are sector publications, not an independent sample of all contemporary expectations. Their positive framing is relevant as attributed commentary, not proof that their authors or Deloitte ignored adverse evidence. No quantitative sentiment score or causal attribution is calculated.

Dates, vintages and stopping rule

The press pages explicitly state 6 April 2021 and 24 March 2022. Their reported sales periods end on 27 December 2020 and 26 December 2021. The retained HTML was captured on 19 October 2026; its hash identifies that capture, not necessarily the wording visible on the original publication day. PDF upload paths likewise do not establish publication dates. The Indian economic summary's precise publication date remains unresolved.

Do not subtract market levels from separate releases to reconstruct an annual trend: their category coding, coverage or prior-year restatements may differ. The bounded comparison uses the growth and category conditions reported within each dated release.

The substantive test stops at the two contemporary dated press releases and the Indian economic summary. Later industry-report passages were consulted during local review but are excluded from this argument, because subsequent outcomes cannot establish what was known in 2021–22. No broad sentiment search, new model run or fresh source retrieval was performed. Further causal attribution would require a dated model-development record and explicit parameter rationale from GFI India/Deloitte. Without it, report the mechanically demonstrated parameter-bound difference separately from this hypothesis.

Suggested report wording

> Earlier sector expectations may be part of the context, but we cannot establish that they caused GFI India's higher scenarios. GFI's 2021–22 US releases were upbeat, yet the March 2022 release already reported much slower aggregate growth than the preceding release and flat plant-based meat dollar sales, alongside inflation and supply disruption. The clearer documented difference is methodological: GFI describes category-level adoption, prices and supply-side assumptions, while our aggregate proxy has chosen growth bounds and no separately identified price-parity or category-entry effects. Its residual growth term may reflect some of those mechanisms; their individual contributions cannot be recovered from this model.

The proposed wording is an author synthesis, not a quotation from GFI or a statement of reviewer acceptance.

Measurement contract and proposed harmonization

Proposed for review, 22 October 2026. No team agreement is asserted.

Strand	Outcome / channel	Technology and category	Geography / time	Uncertainty meaning
Regional proxy	Intended domestic retail spending; the Indian source's retail basis is assumed	Plant-based replacement foods plus fermentation-derived human foods; no cultivated meat; aggregate anchor limits category detail	India/Pakistan; 2022 Indian estimate converted to 2025 prices; 2026/2030/2035 scenarios	Percentiles under analyst-selected parameter distributions; no demonstrated coverage
Regional panel	Headline retail spending; excludes foodservice and conventional staples	Plant-based replacements, all dairy alternatives and other included plant-based replacements; headline excludes fermentation/cultivated technologies	India/Pakistan; 2026/2030/2035; million market-exchange-rate USD at 2025 prices	Percentiles of ten accepted response medians by requested model; not predictive bounds
Separate panel US arm	Original all-category plant-based retail question; additional technology questions have their own scope	Broader US plant-based basket, with distinct cultivated and fermentation questions	US methodological comparator in the author's supplement	Same descriptive run-median aggregation; prompts and supplied evidence differ by region
Historical US benchmark	GFI/SPINS observed retail sales	Broad basket including prepared meals	Fixed 2025 observation; per-person denominator	Source estimate; not a 2035 US forecast
Conventional food evidence	Supply balance or reported household food quantities	Animal foods and conventional categories	Source-specific year, sample and geography	Measurement/sampling properties; does not establish alternative-protein purchases

For any other contribution to the unified report, record the actual executed question, product basket, channel, currency, price year, source vintage and uncertainty measure before comparing it. A common label introduced after collection does not make different protocols equivalent.

The main research question is what the evidence supports about demand and what would reduce decision-relevant uncertainty. Keep stated willingness, paid first purchase, repeat purchase and animal-food displacement separate. Model output, policy intent, company activity and missing data are different evidence classes. Do not sum regional results into a global market, pool proxy draws with model responses, or attribute disagreement to one cause without a design that identifies it.

Separate forecast-elicitation supplement

AI-assisted methods draft, 22 October 2026. Human verification pending. Authored for Muhammad Umar Zafar. This is a separate protocol and sample, not an accepted joint result.

1. Purpose and limits

The exercise records repeated model-generated projections under specified prompts and execution conditions. It can describe within-workflow repetition and between-workflow disagreement. It cannot establish forecast calibration, independent expert opinion, causal effects of AI on food demand, or why a provider's outputs differ from another's.

The approved target was ten accepted responses per requested model and region: two models across India, Pakistan and the US, giving six cells and 60 accepted responses. All registered cells reached ten accepted responses. "Complete" describes that target, not scientific acceptance.

2. Instrument and country scope

The US questionnaire preserves the author-designated source text received for this execution. The source DOCX and derived text hashes are retained. Byte agreement proves provenance of this supplied version; it does not prove that every run in another study used exactly that prompt, background material or software. The original instrument and regional adaptations are local private inputs because public redistribution rights are not established.

Arm	Outcome and product scope	Conditioning
India/Pakistan	Retail plant-based meat/seafood replacements; all dairy alternatives; other in-scope plant-based replacement products. Headline excludes fermentation and cultivated technologies, conventional staples and foodservice.	Country-adapted questions plus frozen public evidence summaries.
US	Original broad all-category plant-based retail question. Separate cultivated and fermentation questions retain their original retail/foodservice and pet-food inclusions.	Original questionnaire; no added project market figures in the US evidence pack.

Forecast years are 2026, 2030 and 2035. Headline unit is **million USD at constant 2025 prices**. Regional category medians must sum to headline medians; marginal tail quantiles are not required to sum. All dairy alternatives in the questionnaire are broader than the milk-only calibration used in a separate household simulator. Results must not be relabelled as milk-only or as that simulator's validated inputs.

The regional instrument also requests stated-intent conversion, Bass coefficients, price-parity conditions, AI-attributed manufacturing-cost decline and separately defined trade/listing/external-estimate proxies. Those variables have different units and resolution conditions; none is a substitute for measured headline

retail demand. The India/Pakistan national headline is presently unresolvable from a complete observed series. A null parity year denotes the stated unknown/never condition, not zero. AI manufacturing-cost decline is not retail pass-through. A “first legal sale” question is distinct from a sales total.

3. Evidence selection and possible anchoring

Country evidence packs were frozen on 19 October 2026. India includes the external GFI/Deloitte estimate and forward scenarios, conventional household consumption context, sample-specific stated-intent research, trade-proxy limitations and policy context. Pakistan includes conventional household context, an upstream trade observation, the same non-Pakistan intent comparison, and explicitly identified India-only external estimates. Neither pack includes project household-model results, internal price ratios, project market forecasts or unit-record data.

The packets themselves disclose that some cited sources rely on earlier catalogue review rather than a fresh freeze-day retrieval. An empty Comtrade capture means missing data, not zero trade. Upstream protein concentrates and listing counts are not alternative-protein retail sales. The GFI forward scenarios may anchor respondents; their broader technology scope differs from the regional headline.

The US evidence file does not receive the added regional facts. Therefore “all runs had the same evidence pack” is false across regions. Similar isolation rules also do not establish equal executed retrieval. Evidence-conditioned elicitation differs from a blank-slate protocol that omits supplied historical observations and external forecasts. This exercise does not isolate that effect and is not a controlled experiment on evidence provision.

4. Execution, model identity and dates

Requested model labels were `gpt-6-astra` and `claude-opus-5-5`. Every accepted Claude receipt reports `claude-opus-5-5`. Accepted Astra receipts have an empty `reported_models` field; their identity is documented as requested, **not independently confirmed by provider-returned model metadata**. Different service names do not establish independent architectures or independent evidence.

Requested model	Accepted-run time window, UTC	Accepted cells
gpt-6-astra	19 October 2026, 18:40:58–19:50:24	Ten India, ten Pakistan, ten US
claude-opus-5-5	20 October 2026, 19:15:23 to 21 October 2026, 15:12:55	Ten India, ten Pakistan, ten US

Each attempt uses a fresh temporary working directory, no resumed conversation and no prior model answers. Claude is configured with WebSearch/WebFetch, restricted mode, an explicit system instruction, no user/project settings and an empty strict MCP configuration. Codex is configured to disable shell, project instructions, memories, apps, skill discovery and delegation, and to use read-only sandboxing with provider-event auditing. The runner documents that Codex has no demonstrated blanket end-user tool allowlist; observed prohibited command/file/MCP/delegation use is rejected. These are implementation and configuration claims, not proof that every possible contamination route was absent.

The retained receipt audit establishes configured web access for completed recorded attempts. It does **not** establish that all runs executed the same searches, fetched the same pages or had equivalent tool behavior. Search parity requires a separate event-level extraction and is not asserted here. Prompts, provider events and answers stay private; their hashes and sanitized metadata can support integrity checks without redistributing their contents.

5. Attempts, exclusions and protocol change

The receipt inventory contains **79 recorded attempts**, not 79 demonstrated completed model calls:

Requested model / region	Accepted	Rejected	Execution error	Incomplete RUNNING record
Astra / India	10	0	0	0
Astra / Pakistan	10	0	0	0
Astra / US	10	0	0	0
Claude / India	10	7	1	0
Claude / Pakistan	10	0	1	0
Claude / US	10	8	1	1
Total	**60**	**15**	**3**	**1**

Seven rejected Claude India receipts cite a first-sale-year validation rule. Six Claude US receipts cite an unapproved reported model, and two cite JSON parsing. The three execution errors are retained separately. The RUNNING record lacks a command receipt; its existence alone establishes neither a sent call nor completion. No attempt is retroactively renamed accepted to make the design look cleaner. `data/panel_attempts.csv` contains sanitized record states, model identities, dates, reason categories and receipt/parsed hashes.

On 20 October, a parser amendment allowed a cultivated first-sale year accompanying a “no” answer to be normalized to null. The original supplied year is retained as `cultivated_conditional_year_dropped`. Before that amendment, seven of eleven Claude India attempts had been rejected for this condition alone, compared with zero of the thirty Astra attempts. Previously rejected receipts remained rejected. Ten accepted responses per cell therefore do not imply equal selection mechanisms. The amendment, failures and retries may affect the accepted distributions; this supplement does not estimate that selection effect.

The runner validates units, finite values, year coverage, monotone quantiles, category coherence and prohibited source/tool contamination. It retains failures and uses fresh retries. Configuration defaults include bounded validation attempts and a 900-second timeout; actual slot histories can exceed one default batch through continuation. The attempt table, not a default-limit description, is the executed count record. No rerun was performed for this drafting task.

6. Aggregation and retained results

For each model, region, measure and year, the summary takes each accepted response's stated P50 and computes empirical percentiles across those run medians. It does **not** pool all response P10/P50/P90 values into one predictive distribution. Original quantiles are retained in the numeric response archive. Missing conditional values are excluded for that measure and `n_nonmissing` is reported; missingness is not filled with zero.

Sort the run medians as `x[0]…x[n−1]`. For fraction `q`, let `h=(n−1)q`, `i=floor(h)`, `j=min(i+1,n−1)`; return `x[i]+(x[j]−x[i])(h−i)`. For ten nonmissing runs, the median is the mean of the fifth and sixth ordered values, P10 is `0.1×first + 0.9×second`, and P90 is `0.9×ninth + 0.1×tenth`.

For a purely illustrative ordered set of ten run medians `1,2,…,10`, this rule gives P10=1.9, P50=5.5 and P90=9.1. These example numbers explain the interpolation; they are not study results or a model's own predictive quantiles.

Region	Requested model	2035 P10 of run medians	Median of run medians	P90 of run medians
India	Astra	324	370	422
India	Claude	84.2	157.5	190.7
Pakistan	Astra	27.6	37	60
Pakistan	Claude	7.96	9.35	15.32
US	Astra	10,500	10,650	11,000
US	Claude	6,880	7,000	7,320

All values are million USD at 2025 prices; each cell has ten accepted, nonmissing headline medians. These are descriptive workflow outputs, **not calibrated predictive intervals**. No provider weights or grand pooled median are introduced. The India/Pakistan panel excludes fermentation whereas the proxy model includes it; their differences cannot be attributed solely to model optimism or evidence conditioning. The US arm remains in this separately authored methods supplement and does not expand Umar's main regional chapter into a US research claim.

7. Relationship to other forecasting studies

Any comparison requires the other study's executed-method record and author approval. Product scope, prompts, supplied evidence, execution date, tool use, model identity, number of submissions and aggregation can all differ. An envelope formed from minimum P10 and maximum P90 is not the same statistic as percentiles of repeated-run medians. An internally aggregated submission must not be counted as several equivalent observed runs.

A terminology decision after execution cannot change those differences. Discuss compatible summaries side by side with their definitions; do not pool them into a larger supposedly independent panel. This package contains no collaborator draft, private result table or claim of another author's approval.

8. Reproduction, rights and acceptance

Run `python scripts/reproduce.py` from the package root to reconstruct the **headline table** from `data/panel_runs.json`. The packaged `scripts/build_panel.py` implements the linear quantile rule above; it checks all six ten-run cells and rejects invalid units, duplicate slots and non-finite, negative or crossed quantiles. The reproduced 18 summaries are compared with `data/reference_panel_aggregates.json` at absolute tolerance 1e-12. This route uses only the Python standard library and no private answers, questionnaire text, provider logs, credentials or new model calls.

The complete elicitation is not independently rerunnable from a public package while questionnaire rights remain unresolved. An integrity hash does not grant redistribution rights. Retain original records privately and state this reproducibility boundary. Publicly proposed rows and metadata still require a content/rights review and human verification.

No other author's acceptance or agreement is claimed. Human factual/code review, author approval and source-specific release review are outstanding. Joint inclusion requires a separate editorial decision. Source pointers and hashes are recorded in the companion claim map and provenance files.

Limitations and AI-use statements

Local review candidate, 27 October 2026. Human verification pending.

Main-text limitations

The India–Pakistan results are conditional spending scenarios. The Indian anchor is an external modeled estimate with an unstated price and valuation basis; the analysis assumes nominal 2022 rupees and retail value. Pakistan lacks a comparable verified national baseline in the retained evidence. Proxy coefficients and distributions are analyst choices, with possible overlap among income, urbanisation, animal-protein supply and industry activity. The reported simulation percentiles have no demonstrated probability coverage. Macro projections beyond 2031 extend the last available growth rates. The US comparison uses a different product basket and a 2025 observation rather than a matched forecast-year benchmark. National supply, stated intention, paid purchase, repeat purchase and animal-food displacement remain different outcomes. The source collection is selective and does not establish that evidence absent from this collection is absent everywhere.

The separate model-elicitation supplement is a repetition and disagreement exercise under recorded workflows. Its regional prompts, evidence conditioning, execution dates, tools and technology scope can differ from other contributors' studies. Model versions and attempted-run selection also matter. The accepted set excludes rejected and errored attempts, and a parser rule changed during collection. The spread of run medians is not a predictive interval. Similar answers do not prove calibration or independent evidence; different answers do not isolate the effect of a model, prompt or AI assumption. The supplement is authored as Umar's work and is not represented as an accepted joint result.

Limits of the GFI comparison and revised recommendations

The GFI comparison is an inverse-model diagnostic under an unresolved source price basis. Original rupee values accompany conditional dollar conversions. A dash in GFI's fermentation baseline remains undefined, not an observed zero. Mechanical factorization does not identify causal contributions to the disagreement. The chosen model bounds can preclude GFI's targets; that alone does not make either forecast empirically correct. GFI supplied the proxy baseline and forward scenarios in the regional elicitation packs, so these strands are not independent validation.

Changing the US-derived growth floor or fixing elasticity probes sensitivity; neither is an estimated South Asian correction. Pakistan remains a transfer without a verified national sales anchor. A driver with no direct national evidence is labelled as such. Proposed price, repeat-purchase, certification, technical and availability tests have no established impact or funding ranking. The original US panel's three themes remain reviewer-supplied comparison prompts until its actual executed study and author scope are verified; they are not pooled with the separate 60-run supplement.

Current, truthful AI-use statement

AI tools assisted source discovery and summarization, code development, quantitative scenario construction, structured forecast elicitation, and drafting/editing. The separate elicitation requested `gpt-6-astra` and `claude-opus-5-5`; provider identity evidence is reported in its methods rather than assumed.

Codex assisted the 22 October writing and methodological audit and the 27 October revision, including numerical companion development, source inspection, driver/recommendation synthesis, narrative editing and local document/browser verification. The revision did not collect new elicited forecasts or change the retained historical model outputs. The human author's scope choices and approvals are recorded separately from factual verification. Human verification of the included claims, citations, statistics, quotations, datasets and code is pending. These materials are review drafts and must not be represented as final or accepted programme outputs.

This statement describes known workflow categories. Before final submission, the human author must complete a tool/version/task register for substantive earlier AI use as well; this drafting pass cannot certify the full historical record.

Final statement template — do not use until completed

> AI assistance was used for [verified list of tools and substantive tasks]. [Named humans] checked [scope and dated review records], including the cited originals, numerical transformations and executable code. Remaining limitations are reported in [locations]. [Insert the programme's required attestation only after it is true.]

The programme policy requires human verification of **all** included AI-produced or AI-edited claims, citations, statistics, quotations, datasets and code. Prioritize load-bearing claims, but do not narrow the requirement to those claims alone. Code review includes reading, running and testing. Agent checks do not substitute for human review. Scholarly peer review may remain pending and labelled after factual/code verification; that label is not an alternative to verification.

Private review drafts may retain explicitly marked unverified material for review. Final submission/public release must omit it or complete the verification. Source: Sentient Futures Project Incubator AI Use Policy, 20 August 2026, Verification, Transparency and Attestation sections. The policy document is not bundled; participants should consult their programme copy.

Human review register fields

Claim ID; exact manuscript sentence/table; original-source locator; source vintage; numerical transformation/code revision; reviewer name; review date; disposition (verified/corrected/omit/unresolved); restrictions; author sign-off. A blank reviewer field remains pending. No reviewer identity or approval is inferred from possession of a file.

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1. Driver evidence and proposed tests, evidence ledger, and recommendation register. Sources are selective and country- or product-specific; proposals remain untested. No actor, partner or budget is committed.↩
2. Christopher Bryant et al., "A Survey of Consumer Perceptions of Plant-Based and Clean Meat in the USA, India, and China," *Frontiers in Sustainable Food Systems* 3 (2019), doi:10.3389/fsufs.2019.00011. Affordability figures: FAO and partners, *The State of Food Security and Nutrition in the World 2025*, Annex A1.6, 2024 values; see drivers. Claims C11 to C12. The nine-study extraction package is selective, not a systematic national review.↩
3. Amrutha Girivasan and Chandana Tekkatte, GFI India, *A comprehensive analysis of India's talent landscape for the smart protein sector*, physical PDF page 10. The official website listing is dated 19 December 2025; the PDF itself is undated. AltProtein.Jobs extracts and the author's capacity proposals are claims C16 to C17. Pakistan's proposed pilot is a recommendation, not an observed programme.↩

4. Government of India, BioE3 Cabinet approval, 24 August 2024; Pakistan, *Pakistan Halal Authority Act, 2016*, especially establishment and functions. These are dated institutional sources, not a legal determination for a particular product. Claims C13 to C14.↩
5. GFI India and Deloitte, *Smart Protein Economic Analysis*, 2022-context analysis, methodology and final-page domestic-market table, original report. Precise publication date and price base are unstated; the upload path is not a publication date. Claims C01 to C03 in the claim map.↩
6. Model inputs and code: `altprotein_proxy_inputs.csv`, `altprotein_proxy_scenarios.csv` and `build_altprotein_proxy.py`; exact equations and assumptions in proxy methods. IMF WEO October 2026 and World Bank WDI provide macro inputs. Claims C04 to C06.↩
7. Sensitivity rows `market_value_drift_overlap_adjusted` and `market_value_without_urban_animal` in the reference scenarios; the 2030 overlap value is in sensitivity-summary.csv. See methods for the exact changes. Claims C09 to C10.↩
8. Reproducible companion, GFI gap methods, diagnostic tables, and source category extraction. All converted comparisons keep the unresolved GFI price-basis assumption; reference headline outputs are unchanged.↩
9. Reproducible companion, GFI gap methods, diagnostic tables, and source category extraction. All converted comparisons keep the unresolved GFI price-basis assumption; reference headline outputs are unchanged.↩
10. GFI US retail market data, 2025 sales; IMF population. The US anchor is $7.9 billion, or $23.106847 per person. Claims C07 to C08.↩
11. Reproducible companion, GFI gap methods, diagnostic tables, and source category extraction. All converted comparisons keep the unresolved GFI price-basis assumption; reference headline outputs are unchanged.↩
12. Separate elicitation method, especially scope, evidence conditioning and aggregation. Sixty accepted responses from 79 recorded attempts; no new elicitation in this revision.↩
13. Driver evidence and proposed tests, evidence ledger, and recommendation register. Sources are selective and country- or product-specific; proposals remain untested. No actor, partner or budget is committed.↩
14. AI-mechanism source review, including Cosenza et al., doi:10.1002/elsc.202300005, and Watson et al., doi:10.1038/s41586-023-06415-8. The former's primary full text was not freshly recovered for this drafting pass. Claim C15.↩
15. GFI India and Deloitte India, *Smart Protein Economic Analysis*, physical PDF page 1 (Methodology), page 2 (scenario explanation), and page 4 (domestic table). The 2022 context and original reported values are retained; the URL upload path is not treated as a publication date.↩
16. GFI India and Deloitte India, *Smart Protein Economic Analysis*, physical PDF page 1 (Methodology), page 2 (scenario explanation), and page 4 (domestic table). The 2022 context and original reported values are retained; the URL upload path is not treated as a publication date.↩
17. IMF, *World Economic Outlook database*, October 2026, LP and NGDP_RPCH series. Official DataMapper source. Retained source ID `imf_weo_2026_10`; see proxy methods §3.2 for extrapolation beyond 2031.↩
18. FAO, IFAD, UNICEF, WFP and WHO, *The State of Food Security and Nutrition in the World 2025*, Annex A1.6, physical PDF page 176, India and Pakistan, 2024 percentage column. Original report. Source ID `fao_sofi_2025`.↩
19. Christopher Bryant et al., "A Survey of Consumer Perceptions of Plant-Based and Clean Meat in the USA, India, and China," *Frontiers in Sustainable Food Systems* 3 (2019), physical PDF pages 1–3, abstract and methods. Original article. Source ID `india_cultural_bryant_2019`.↩
20. GFI India and Deloitte India, *Smart Protein Economic Analysis*, physical PDF page 1, Methodology, and page 4, domestic table. Original report. Source ID `gfi_india_smart_protein_economic_analysis_2022`; exact publication date and valuation basis remain unestablished.↩
21. FAO, FAOSTAT Food Balances, 28 October 2025 vintage, 2023 observations. Source ID `fao_fbs_2025_10_28`; seven animal-source protein-supply inputs are identified in proxy methods §3.4.↩
22. Christopher Bryant et al., "A Survey of Consumer Perceptions of Plant-Based and Clean Meat in the USA, India, and China," *Frontiers in Sustainable Food Systems* 3 (2019), physical PDF pages 1–3, abstract and methods. Original article. Source ID `india_cultural_bryant_2019`.↩
23. *Pakistan Halal Authority Act, 2016*, Act VIII of 2016, §§1, 3 and 10. Dated institutional text. Source ID `pakistan_cultural_halal_act_2016`. This reference is not an exhaustive current legal or commencement assessment.↩
24. Government of India, Press Information Bureau, "Cabinet approves BioE3 policy," 24 August 2024, paragraphs on R&D, Bio-AI infrastructure and smart proteins. Official announcement. Source ID `india_pib_bioe3_2024`.↩
25. Amrutha Girivasan and Chandana Tekkatte, GFI India, *A comprehensive analysis of India's talent landscape for the smart protein sector*, physical PDF page 10, industry and academic cohorts; page 12, technical skill gaps. Official source. The listing is dated 19 December 2025; the PDF is undated. Source ID `gfi_india_talent_landscape`; 20 industry and 19 academic respondents are cohorts, not workforce totals.↩
26. Retained aggregate inputs in `altprotein_proxy_inputs.csv`, source IDs `altprotein_jobs_vacancy_2023_2025` and `altprotein_jobs_company_directory_2024`; proxy methods §3.3. These records do not establish a national company census or independently verified board coverage. Original workbooks are not redistributed.↩
27. Government of India, Press Information Bureau, "Cabinet approves BioE3 policy," 24 August 2024, paragraphs on R&D, Bio-AI infrastructure and smart proteins. Official announcement. Source ID `india_pib_bioe3_2024`.↩
28. Retained proxy specification, §§1, 3.3–3.4 and 6. Model-specification and missing-evidence statements are bounded to this package, not claims that no relevant evidence exists elsewhere.↩

29. GFI/PBFA, "Plant-based food retail sales grow 27 percent to reach $7 billion in 2020", 6 April 2021, dated opening, pandemic/context discussion and methodology. Source ID `gfi_us_sales_press_2021`.↩
30. GFI/PBFA, "U.S. plant-based food retail sales hit $7.4 billion", 24 March 2022, opening, plant-based meat paragraph, macro/price discussion and methodology. Source ID `gfi_us_sales_press_2022`.↩
31. GFI India/Deloitte India, *Smart Protein Economic Analysis*, physical PDF page 1, Methodology, and page 4, domestic scenarios. Source ID `gfi_india_smart_protein_economic_analysis_2022`. Exact publication date and price/retail basis remain unresolved.↩